



OBJECT-ORIENTED PROGRAMMING

THE SOFT WARRIORS

REQUIREMENTS FEATURES

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SANGOLQUÍ-ECUADOR

1. **Backup System**

The system shall allow the store manager to create and restore backups of all system data, upload backups to the cloud, and download backups from the cloud for data recovery.

2. **Cash Control**

The system shall allow the cashier to manage the cash register by recording the starting balance, security withdrawals, and total sales. The system shall require the cashier to enter the physical cash count before displaying the expected balance, automatically recording discrepancies and partial withdrawals to prevent cash leakage.

3. **Combos**

The system shall allow the store manager to create promotional bundles, for example a breakfast kit that contains bread, milk and eggs. The system needs to deduct stock for each item individually and validates that the final price maintains a positive profit.

4. **Credit Management**

The system shall identify customers who have stopped purchasing staple products for a defined period and suggest personalized offers to the store manager to recover their loyalty.

5. **Expiration Control**

The system shall generate expiration alerts for products nearing their expiration date and automatically calculate progressive discounts to encourage quick stock rotation.

6. **Slow Moving Report**

The system shall analyze products with low turnover rates and generate reports to identify slow-moving inventory, helping the manager make informed decisions about discounts or discontinuing products.

7. **Stock Min Alert**

The system shall allow the store manager to calculate the minimum critical stock levels based on sales, generating a suggested order list organized by supplier before a product runs out.

8. **UI Adaptive**

The system shall allow users to customize the interface appearance, including theme selection, font size adjustment, high contrast mode, and keyboard shortcuts for accessibility.

9. **Product Management**

The system shall allow the store manager to manage the complete product inventory, including adding new products, updating product information, searching by ID or name, deleting products, and controlling stock levels. All product data shall be persisted in products.json with support for differentiated wholesale and retail pricing.

10. **Sales System**

The system shall allow the cashier to manage sales transactions by recording the sale ID, product list, total amount, and payment type (cash, credit, or mixed). The system shall classify sales according to applicable tax rates (15% IVA) and generate monthly fiscal summaries. The system shall allow the cashier to place a sale on hold when a customer forgets an item or payment method, enabling service to continue for the next customer without losing the original cart data. The system shall also analyze the current shopping cart and suggest complementary products

based on local purchasing habits to support cross-selling opportunities.

11. Supplier Management

The system shall allow the store manager to track returnable containers between the customer, the store and the suppliers, recording deposits or exchanges as part of the payment process. The system shall allow the store manager to enter supplier invoices, detect increases in supplier costs, and automatically suggest a new retail price to maintain business profitability. The system shall also allow the store manager to manage supplier credit terms and compare historical prices to alert when a supplier is charging above the usual market average.

12. Reservations

The system shall allow the store manager to make reservations for clients that require so, the product need to be removed from the available stock. The reservation has an expiration date and can be completed, cancelled, or extended.